

Preconditioners for Singular Black Box Matrices

William J. Turner

Department of Mathematics & Computer Science
Wabash College
Crawfordsville, IN 47933

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Outline

Generalized Networks

- Arbitrary Radix Switching Networks

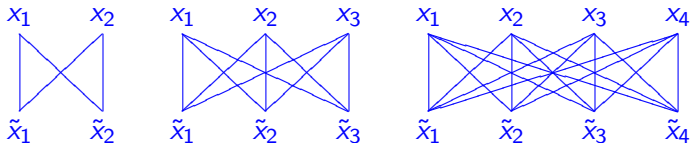
- Generic Exchange Matrices

Rank Preconditioner

Arbitrary Radix Switches

Generalize butterfly switch to radix- ρ switch:

Example (Radix- ρ switches: $\rho = 2, 3, 4$)

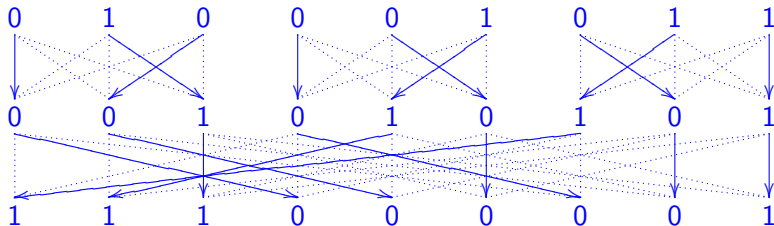


Arbitrary Radix Switching Networks

Definition (ℓ -dimensional radix- ρ switching network)

- ▶ Recursively defined
- ▶ ρ^ℓ radix- ρ switches to merge outputs of ρ radix- ρ subnetworks of dimension $\ell - 1$
- ▶ Merges i th output of each of the subnetworks

Example ($\rho = 3$, $\ell = 2$)

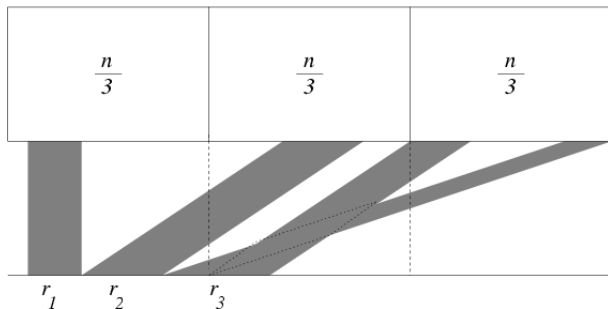


Arbitrary Radix Switching Networks

Lemma

Let $n = \rho^\ell$ where $\rho \geq 2$. The ℓ -dimensional radix- ρ switching network can switch any r indices $1 \leq i_1 < \dots < i_r \leq n$ into any desired contiguous block of indices. For our purposes, wrapping the block around the outside preserves contiguity.

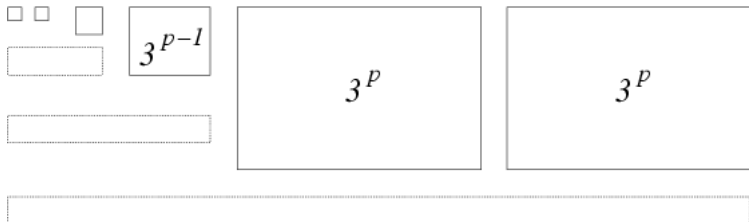
Example ($\rho = 3$)



Generalized Arbitrary Radix Switching Networks

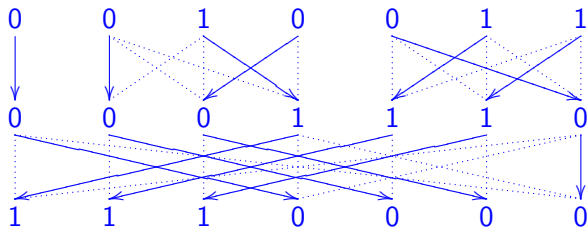
Definition (Generalized radix- ρ switching network)

- ▶ Let $n = \sum_{i=1}^p n_i$ where $\begin{cases} n_i = c_i \rho^{\ell_i} \\ c_i \in \{1, \dots, \rho - 1\} \\ \ell_1 < \ell_2 < \dots < \ell_p \end{cases}$
- ▶ Radix- $(c_k + 1)$ switches to merge $\sum_{i=1}^{k-1} n_i$ subnetwork with far right nodes of ρ^k blocks
- ▶ Radix- c_k switches to merge other nodes of ρ^k blocks



Generalized Arbitrary Radix Switching Networks

Example ($\rho = 3, n = 7 = 1 \rho^0 + 2 \rho^1$)



Theorem

Suppose $\rho \geq 2$. The generalized radix- ρ switching network described above can switch any r indices $1 \leq i_1 < \dots < i_r \leq n$ into the contiguous block $1, 2, \dots, r$. Furthermore, it has a depth of $\lceil \log_\rho(n) \rceil$ and a total of no more than $\rho^{\lceil \log_\rho(n) \rceil} \lceil \log_\rho(n) \rceil / \rho$ switches of radix at most ρ . The network attains this bound only when $n = \rho^{\lceil \log_\rho(n) \rceil}$ or $n < \rho$.

Building Preconditioners

- ▶ Replace each switch by $\rho \times \rho$ symbolic exchange matrix S_k to merge columns (not mix)
- ▶ Embed each exchange matrix as principal minor of $n \times n$ identity matrix to create $\hat{S}_k$
- ▶ Symbolic preconditioner is product: $\mathcal{L} = \prod_{k=1}^s S_k$
- ▶ Randomly choose values for symbols to create probabilistic preconditioner: $\tilde{A} = AL$ where $L = \prod_{k=1}^s S_k$

Building Preconditioners

Example ($\rho = 3$)

$$\mathcal{S}_k = \begin{bmatrix} \alpha_{1,1,k} & \alpha_{1,2,k} & \alpha_{1,3,k} \\ \alpha_{2,1,k} & \alpha_{2,2,k} & \alpha_{2,3,k} \\ \alpha_{3,1,k} & \alpha_{3,2,k} & \alpha_{3,3,k} \end{bmatrix}$$

$$\hat{\mathcal{S}}_k = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & \alpha_{1,1,k} & \alpha_{1,2,k} & \alpha_{1,3,k} & \\ & & & \ddots & & \\ & \alpha_{2,1,k} & \alpha_{2,2,k} & & \alpha_{2,3,k} & \\ & & & & \ddots & \\ \alpha_{3,1,k} & \alpha_{3,2,k} & & & \alpha_{3,3,k} & \\ & & & & & \ddots \\ & & & & & & 1 \end{bmatrix}$$

Symbolic Toeplitz Matrix

$$\mathcal{T} = \begin{bmatrix} \alpha_\rho & \alpha_{\rho+1} & \dots & \alpha_{2\rho-1} \\ \alpha_{\rho-1} & \alpha_\rho & \dots & \alpha_{2\rho-2} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1 & \alpha_2 & \dots & \alpha_\rho \end{bmatrix}$$

Using the lexicographic monomial ordering with

$$\alpha_1 \prec \alpha_2 \prec \dots \prec \alpha_{2n-1}:$$

$$\text{lt} \left(\det \left(\mathcal{T}_{[i_1, \dots, i_m; j_1, \dots, j_m]} \right) \right) = (-1)^{\lfloor m/2 \rfloor} \prod_{k=1}^m \alpha_{n+j_{m+1-k}-i_k}$$

Toeplitz Exchange Matrix

Let A' be submatrix of A affected by T .

- ▶ A' includes a subset of r' of the r linearly independent columns of A where $0 \leq r' \leq r$.

- ▶
$$\det((A'T)_{[\mathcal{I}, \mathcal{J}]}) = \sum_{\substack{\mathcal{K} = \{k_1, \dots, k_m\} \\ 1 \leq k_1 < \dots < k_m \leq n}} \det(A'_{[\mathcal{I}, \mathcal{K}]}) \det(T_{[\mathcal{K}, \mathcal{J}]})$$

- ▶ Every set of $m \leq r'$ columns of $A'T$ linearly independent
- ▶ Embedded Toeplitz exchange matrix $\hat{T}$ can move linear independence freely within one switch.
- ▶ Each $\hat{T}_k$ uses different symbols $\implies$ first r columns of $A(\prod \hat{T}_k)$ linearly independent.

Toeplitz Exchange Matrix

Theorem

Let $\mathbb{F}$ be a field, let $A \in \mathbb{F}^{n \times n}$ have r linearly independent columns, let s be the number of switches in the generalized radix- ρ switching network, and let S be a finite subset of $\mathbb{F}$. Let N be the number of random numbers required in this network. If $a_1, \dots, a_N$ are randomly chosen uniformly and independently from S , then the first r columns of $A(\prod_{k=1}^s \hat{T}_k)$ are linearly independent with probability at least

$$1 - \frac{r \lceil \log_{\rho}(n) \rceil}{|S|} \geq 1 - \frac{n \lceil \log_{\rho}(n) \rceil}{|S|}.$$

Toeplitz Rank Preconditioner

- ▶ $\rho = n \implies$ any r columns of AT linearly independent
- ▶ Some $r \times r$ principal minor of AT nonzero
- ▶ ATD preconditioner for Kaltofen-Saunders rank algorithm [Turner(2002), Ch.3]
- ▶ Do not need D !

Characteristic Polynomial

- ▶ $\det(\lambda I - AT) = \sum_{m=0}^n (-1)^m E_m(AT) \lambda^{n-m}$
- ▶ $E_m(AT) = \sum_{\substack{\mathcal{J}=\{i_1,\dots,i_m\} \\ 1 \leq i_1 < \dots < i_m \leq n}} \det((AT)_{[\mathcal{J},\mathcal{J}]})$
- ▶ $m > r \implies E_m(AT) = 0$
- ▶ λ^{n-r} divides $\det(\lambda I - AT)$

Leading Term of Characteristic Polynomial

Let

- ▶ $\mathcal{P} = \{p_1, \dots, p_r\}$ where $1 \leq p_1 < \dots < p_r \leq n$ be the pivot column indices of A
- ▶ $\mathcal{Q} = \{q_1, \dots, q_r\}$ where $1 \leq q_1 < \dots < q_r \leq n$ be the indices of the last r linearly independent rows of A

Then, under the lexicographic monomial ordering with

$$\lambda \prec \alpha_1 \prec \alpha_2 \prec \dots \prec \alpha_{2n-1},$$

$$\text{lt}(\det(\lambda I - AT)) = (-1)^{\lfloor 3r/2 \rfloor} A_{[\mathcal{Q}, \mathcal{P}]} \lambda^{n-r} \prod_{k=1}^r \alpha_{n+p_{m+1-k}-q_k}$$

Characteristic Polynomial

- ▶ $\det(\lambda I - AT) = \lambda^{n-r}g$ where $\deg_{\lambda}(g) = r$ and λ does not divide g
- ▶ $\text{lt}(g)$ constant in λ and at most linear in each α_i
- ▶ g squarefree
- ▶ $f^{AT} = \lambda^p g$ where $p \geq 1$
- ▶ $\deg_{\lambda}(f^{AT}) \geq r + 1$

Degree Upper Bound

- ▶ Sylvester's inequality: $\text{rank}(AT) \leq r$
- ▶ λ divides the $n - r$ largest invariant factors $s_k(\lambda I - AT)$
- ▶ $\deg(f^{AT}) \leq r + 1$
- ▶ $\deg(f^{AT}) = r + 1$

Toeplitz Rank Preconditioner

Theorem

Let $\mathbb{F}$ be a field, $A \in \mathbb{F}^{n \times n}$ have rank r with $r \leq n-1$, and S be a finite subset of $\mathbb{F}$. If

$$T = \begin{bmatrix} a_n & a_{n+1} & \cdots & a_{2n-1} \\ a_{n-1} & a_n & \cdots & a_{2n-2} \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & a_n \end{bmatrix} \in \mathbb{F}^{n \times n},$$

where $a_1, \dots, a_{2n-1}$ are chosen uniformly and independently from S , then the matrix AT has characteristic polynomial $\det(\lambda I - AT) = \lambda^{n-r} g(\lambda)$ and minimal polynomial $f^{AT} = \lambda g(\lambda)$ where $g(0) \neq 0$ and $\deg(f^{AT}) = r+1$, all with probability at least

$$1 - \frac{r(r+1)}{2|S|} \geq 1 - \frac{n(n-1)}{2|S|}.$$

Conclusion

- ▶ Generalization of Chen et al. butterfly preconditioners to arbitrary radix switches
 - ▶ Small increase in probability bound for success
 - ▶ $\rho \rightarrow n$ allows search for other rank preconditioners
- ▶ Monomial ordering and leading term arguments
 - ▶ Toeplitz matrices as arbitrary radix exchange matrices
 - ▶ Toeplitz matrix as rank preconditioner
 - ▶ Better probability of success
 - ▶ Asymptotically slower than butterfly preconditioner
 - ▶ Apply to other matrices?

References



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